

Multimodal Polysomnography for Joint Sleep Staging and Respiratory-Event Detection in Subacute Ischemic Stroke

Md Wahiduzzaman Sua, Esm-e Moula Chowdhury Abha, and Md Imtiaj Alam Sajin

Abstract—A clinical polysomnogram records far more than the brain. Alongside the electroencephalogram it captures eye movements, muscle tone, the electrocardiogram, nasal and oral airflow, thoracic and abdominal breathing effort, and blood oxygen saturation. In patients recovering from ischemic stroke, a group in which breathing is disturbed during sleep at high rates, these cardiorespiratory signals carry the disease itself. We build one model that reads the full polysomnogram and produces two clinical outputs at once: the sleep stage of every 30-second epoch and a per-epoch label for respiratory events. The model encodes the neural and the cardiorespiratory signals in two parallel streams, lets the two streams exchange information through cross-modal attention, follows the night with a bidirectional recurrent decoder, and drives a staging head and a respiratory head, with a direct connection that carries the cardiorespiratory signal to the respiratory head so oxygen desaturation reaches the decision at full strength. We benchmark the model against fifteen architectures on iSLEEPS, the first public polysomnography corpus of subacute ischemic stroke (99 patients), all under patient-independent ten-fold cross-validation. The single model stages sleep at 0.721 accuracy with Cohen’s κ of 0.606, which places it among the strongest models on this cohort, and it detects respiratory events at 0.700 area under the ROC curve, a capability none of the staging models possess. The respiratory read-out is grounded in physiology: when we remove the airflow channel from which the events were scored, detection holds at 0.707, because the model reads oxygen desaturation, breathing effort, cardiac variability, and the cortical arousals that follow an obstructed breath. Each modality contributes where its physiology lives. Code and features are released.

Index Terms—Polysomnography, multimodal learning, sleep staging, sleep apnea, ischemic stroke, cross-modal attention, multi-task learning, cardiorespiratory signals.

I. INTRODUCTION

SLEEP is scored in 30-second epochs into five stages, Wake (W), N1, N2, N3, and rapid-eye-movement (REM), following the American Academy of Sleep Medicine (AASM) rules [1]. The overwhelming majority of automated methods read a single electroencephalogram (EEG) channel and ask one question: how accurately can the five stages be classified? Two decades of work on that question have moved from convolutional and recurrent encoders [2]–[4] through attention and sequence models [5]–[7] to large self-supervised EEG foundation models [8]–[11]. On healthy sleepers these methods reach roughly 85% accuracy, and recent reviews argue the field is near a ceiling set by inter-scorer disagreement [12].

The authors are with the Department of Computer Science and Engineering, American International University-Bangladesh, Dhaka, Bangladesh (e-mail: 26-94088-2@student.aiub.edu; 26-94089-2@student.aiub.edu; 26-94090-2@student.aiub.edu).

A clinical polysomnogram (PSG), however, is a recording of the whole sleeping body. It carries the EEG, eye movements, muscle tone, the electrocardiogram (ECG), nasal and oral airflow, thoracic and abdominal breathing effort, and peripheral oxygen saturation (SpO_2). These cardiorespiratory channels are where sleep-disordered breathing writes itself: a paused breath, a chest that keeps straining against a closed airway, a fall in blood oxygen, and the arousal that rescues the breath. A model that reads only the EEG cannot see any of this, and so cannot report on the disorder that most damages a patient’s sleep. We use the full polysomnogram, and we ask a question that the staging race does not: what does a model gain, and on which clinical task, when it reads the whole recording rather than the brain alone.

We study this on iSLEEPS [13], released in 2026 as the first public PSG corpus dedicated to subacute ischemic stroke. The cohort is well matched to the question. Sleep-disordered breathing is common in the weeks after a stroke and is tied to worse recovery [14], [15], and in this corpus the median patient carries a scored respiratory event in 13% of epochs, with 77 of 96 patients above 5% (Fig. 1). The cardiorespiratory signal is therefore abundant and clinically meaningful in exactly the population where breathing matters most. The published staging baselines on this corpus reach 74.7% accuracy for an LSTM, about ten points below the healthy state of the art, and a concurrent study reports that deep networks attend to non-physiological regions on this cohort [16], so the setting also stresses every staging architecture we can bring to it.

We present a two-stream, multi-task model that treats staging and respiratory-event detection as one joint problem, and we place it inside a broad benchmark of fifteen architectures on the same folds. Our contributions are:

- **One model for two clinical read-outs from the full polysomnogram.** It stages sleep at 0.721 accuracy (κ 0.606) and, from the same forward pass, detects respiratory events at 0.700 AUC, with a direct cardiorespiratory path to the respiratory head so oxygen-desaturation cues reach the decision undiminished (Section IV).
- **A broad benchmark on the stroke cohort.** We compare against convolutional, recurrent, transfer-learned, foundation-style, graph, gradient-boosting, and stacking models, all under patient-independent ten-fold evaluation (Table II). Staging saturates near 0.72 to 0.75 across every strong architecture, and the proposed model sits inside that band while adding a respiratory capability the others lack.

- **A respiratory read-out grounded in physiology, not in the labeling signal.** Removing the airflow channel from which the events were scored leaves detection at 0.707 AUC, because the decision rests on oxygen desaturation, breathing effort, cardiac variability, and cortical arousals (Section V-F).

II. RELATED WORK

A. Single-channel deep sleep staging

Automated staging matured on one EEG channel. DeepSleepNet paired convolutional feature extraction with a bidirectional LSTM [2]; SeqSleepNet framed the night as sequence-to-sequence labeling [3]; AttnSleep introduced attention over multi-resolution convolutions [5]; TinySleepNet and U-Time reduced the parameter budget while holding accuracy [4], [7]; and XSleepNet fused raw and spectral views [6]. These models set the accuracy on healthy corpora but depend on the cortical micro-events that stroke disturbs, which is why they transfer poorly to patients [16].

B. Multi-channel and foundation models

Later work reads several PSG channels. U-Sleep stages from arbitrary EEG and EOG derivations across many cohorts [17], and self-supervised EEG foundation models such as LaBraM, CBraMod, EEGPT, and BIOT learn transferable representations from large unlabeled corpora [8]–[11]. SleepFM extends this idea to several PSG modalities [18]. The eye and muscle channels help most with REM and Wake, which a single EEG channel resolves poorly. Reported gains from adding respiratory channels to *staging* are small and cohort-dependent; our benchmark quantifies this directly on a high-burden clinical cohort and separates the staging effect from the respiratory one.

C. Detection of sleep-disordered breathing

A parallel line of work detects apnea and hypopnea from breathing signals. Airflow and effort shape, oximetry, and ECG-derived heart-rate variability are all informative, and oxygen desaturation is a defining marker of an event [19]. These detectors usually run at fine time resolution on the raw breathing waveforms and are trained on that task alone. We instead detect respiratory events on the staging epoch grid, jointly with staging, from a shared representation, so that one model produces both clinical outputs and the two tasks support each other during training.

D. Sleep and breathing after stroke

Sleep is both disturbed by stroke and predictive of recovery, and sleep-disordered breathing is common in the subacute phase and linked to poorer functional outcome [14], [15]. Focal injury also changes the micro-events that scoring relies on, the spindles, slow waves, and muscle atonia [20]–[22]. A model that reports on breathing as well as stage is therefore of direct clinical value in this population.

TABLE I
STAGE DISTRIBUTION AND RESPIRATORY-EVENT PREVALENCE IN THE CORPUS WE MODEL ($N = 99$).

Stage	W	N1	N2	N3	R
Share (%)	26.6	10.2	42.3	8.9	12.1

Respiratory events: 16.0% of epochs positive; median 13% per patient (range 0 to 60.5%); 77 of 96 patients above 5%.

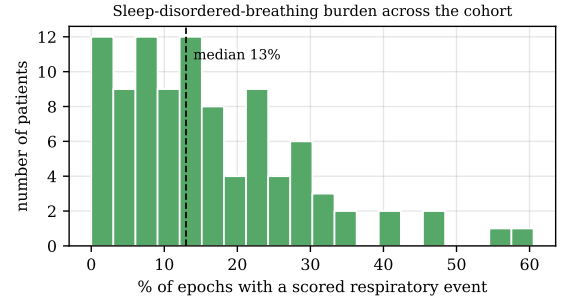


Fig. 1. Sleep-disordered-breathing burden across the cohort. Each bar counts patients by the fraction of epochs that carry a scored respiratory event. The median patient sits at 13% and many exceed 30%, so the cardiorespiratory signal is abundant.

III. DATASET AND PREPROCESSING

iSLEEPS [13] contains overnight PSG from 100 subacute ischemic-stroke patients (mean age 50.5; 77 male, 23 female), scored by experts into the five AASM stages. We exclude one recording after an audit: the data records of SN28 are byte-identical to those of SN15, a duplication consistent with the anonymization and re-numbering the dataset authors describe, and keeping it would place a training subject into the test fold in nine of ten fold assignments. We therefore use $N = 99$ (96 with complete cardiorespiratory channels), a total of 89,532 epochs. The stage distribution (Table I) is strongly imbalanced, with N2 dominant and N1 and N3 scarce, which is the clinical norm.

a) *Channels we read, and channels we do not:* A raw recording holds 22 traces. We read the 14 that carry sleep-stage or breathing information. The *neural stream* takes the four EEG derivations (C4:M1, C3:M2, O2:M1, O1:M2), the two EOG channels (E1:M2, E2:M2), and the submental EMG. The *cardiorespiratory stream* takes the ECG, nasal and oral airflow, thoracic effort, abdominal effort, the summed effort trace, SpO₂, and pulse. We leave out eight traces for stated reasons: periodic limb movement records a different disorder; snore is scored from the same airway sound as the respiratory events and would make the second task circular; plethysmography is already summarized by the SpO₂ and pulse channels we keep; body position is coarse and slow; and the ambient-light and battery traces are device housekeeping, not physiology. All neural channels are resampled to 100 Hz and the cardiorespiratory channels to 25 Hz, and both are read in physical units, which matters because SpO₂ near 90%, pulse near 78 bpm, and microvolt EEG live on very different scales. Per-epoch respiratory labels are the corpus's scored events mapped onto the 30 s grid, and an epoch is positive when it overlaps an

event.

IV. PROPOSED METHOD

A. Problem formulation

A night is a sequence of T epochs. For epoch t we form a neural feature vector $\mathbf{f}_t^{\text{eg}} \in \mathbb{R}^{188}$ and a cardiorespiratory feature vector $\mathbf{f}_t^{\text{car}} \in \mathbb{R}^{14}$, and we learn a subject-independent map

$$f_\theta : (\mathbf{f}_{1:T}^{\text{eg}}, \mathbf{f}_{1:T}^{\text{car}}) \mapsto (\hat{y}_{1:T}^{\text{stg}}, \hat{y}_{1:T}^{\text{apn}}), \quad (1)$$

where $\hat{y}_t^{\text{stg}} \in \Delta^4$ is a distribution over the five stages and $\hat{y}_t^{\text{apn}} \in [0, 1]$ is the probability that epoch t contains a respiratory event (Fig. 2).

B. Physiological feature representations

a) *Neural features* (188): Per channel we take absolute and relative band powers from the Welch spectrum $P(f)$ in delta (0.5–4), theta (4–8), alpha (8–12), sigma (12–16), and beta (16–30) Hz; the relative power in band b is

$$\text{RP}_b = \frac{\int_{f \in b} P(f) df}{\int_{0.5}^{30} P(f) df}. \quad (2)$$

We add spectral entropy $H = -\sum_i p_i \log p_i$ with p_i the normalized spectrum, the 95% spectral-edge frequency, the three Hjorth descriptors [23], and time-domain statistics. Sleep spindles are detected on the sigma-band analytic envelope $e(t) = |\mathcal{H}\{x_\sigma\}(t)|$ from the Hilbert transform $\mathcal{H}$, with density counted above a per-epoch threshold; slow-wave amplitude, ocular-movement energy on the EOG, and tonic EMG level complete the set.

b) *Cardiorespiratory features* (14): From the seven cardiorespiratory channels we compute SpO_2 mean, minimum, variability, and desaturation depth (median minus minimum); pulse mean and variability; ECG amplitude and line-length; airflow amplitude and line-length; thoracic, abdominal, and summed effort amplitudes; and thoraco-abdominal asynchrony, the loss of coordination between chest and abdomen that marks an obstructed breath,

$$\text{asyn}_t = |\text{corr}(x_t^{\text{tho}}, x_t^{\text{abd}})|. \quad (3)$$

Both feature sets are standardized per patient, which removes person-to-person offsets so the model generalizes across people.

C. Two-stream encoders

Each stream is a two-layer perceptron with LayerNorm,

$$e_t = \text{GELU}(\text{LN}(W_2 \text{GELU}(\text{LN}(W_1 \mathbf{f}_t^{\text{eg}})))) \in \mathbb{R}^{128}, \quad (4)$$

$$c_t = \text{GELU}(\text{LN}(U_2 \text{GELU}(\text{LN}(U_1 \mathbf{f}_t^{\text{car}})))) \in \mathbb{R}^{64}. \quad (5)$$

LayerNorm rather than BatchNorm is used because the recurrent windows are short and mix patients within a batch, and the cardiorespiratory stream is deliberately narrower because 14 slow descriptors carry far less structure than 188 neural features.

D. Cross-modal fusion

The two embeddings become two tokens with learned modality-type embeddings $M \in \mathbb{R}^{2 \times D}$, $D = 128$,

$$T = [W_e e_t; W_c c_t] + M \in \mathbb{R}^{2 \times D}. \quad (6)$$

Four-head scaled dot-product attention lets each modality read the other,

$$\text{head}_i = \text{softmax}\left(\frac{(TW_i^Q)(TW_i^K)^\top}{\sqrt{d_k}}\right) TW_i^V, \quad A = [\text{head}_1; \dots; \text{head}_4] W^O \quad (7)$$

followed by a residual connection, normalization, and a projection of the two refined tokens,

$$\tilde{T} = \text{LN}(T + A), \quad z_t = \text{GELU}(W_f \text{vec}(\tilde{T})) \in \mathbb{R}^D. \quad (8)$$

We also evaluate a plain concatenation in place of attention; the two perform alike (Section V-E), so the reported gain is attributable to the modality, not to the fusion mechanism.

E. Temporal decoder and two heads

Sleep is sequential and respiratory events cluster, so the fused vectors of an $L = 20$ epoch window (10 minutes) are read by a two-layer bidirectional LSTM. Each direction runs the standard recurrence

$$\begin{aligned} i_t &= \sigma(W_i[z_t, h_{t-1}]), & f_t &= \sigma(W_f[z_t, h_{t-1}]), \\ o_t &= \sigma(W_o[z_t, h_{t-1}]), & g_t &= \tanh(W_g[z_t, h_{t-1}]), \\ s_t &= f_t \odot s_{t-1} + i_t \odot g_t, & h_t &= o_t \odot \tanh(s_t), \end{aligned} \quad (9)$$

and the forward and backward states are concatenated into $h_t \in \mathbb{R}^{256}$. The staging head is a linear map $\hat{y}_t^{\text{stg}} = \text{softmax}(W_s h_t)$. The respiratory head reads h_t together with a direct copy of the per-epoch cardiorespiratory embedding c_t ,

$$\hat{y}_t^{\text{apn}} = \sigma(W_2 \text{GELU}(W_1[h_t; c_t])). \quad (10)$$

This direct connection matters: without it the respiratory decision must survive a fusion trained mostly for staging, and the fall in oxygen that defines an event is weakened along the way. The full model holds 0.86 million parameters.

F. Training and decoding

The two tasks are trained together with equal weight,

$$\mathcal{L} = - \underbrace{\sum_t \sum_k w_k y_{t,k}^{\text{stg}} \log \hat{y}_{t,k}^{\text{stg}}}_{\text{staging}} + \lambda - \underbrace{\sum_t [\pi a_t \log \hat{y}_t^{\text{apn}} + (1-a_t) \log(1-\hat{y}_t^{\text{apn}})]}_{\text{respiratory}} \quad (11)$$

with $\lambda = 1$. The staging weights use the square root of inverse class frequency, $w_k \propto \sqrt{N/(Kn_k)}$ for $K = 5$ classes with n_k examples, which corrects the scarce N1 and N3 stages without giving up overall accuracy, and the respiratory term is positive-weighted by $\pi = n_-/n_+$ to match the 16% event rate. At inference the staging posteriors are smoothed by a hidden-Markov Viterbi pass whose transition matrix A^H and prior are counted from the training labels [24],

$$\delta_t(j) = \max_i [\delta_{t-1}(i) + \log A_{ij}^H] + \log \hat{y}_t^{\text{stg}}(j), \quad (12)$$

recovered by back-tracking, which restores the natural persistence of sleep. The respiratory head is left unsmoothed. Optimization uses AdamW with cosine annealing and early stopping on a held-out set of patients.

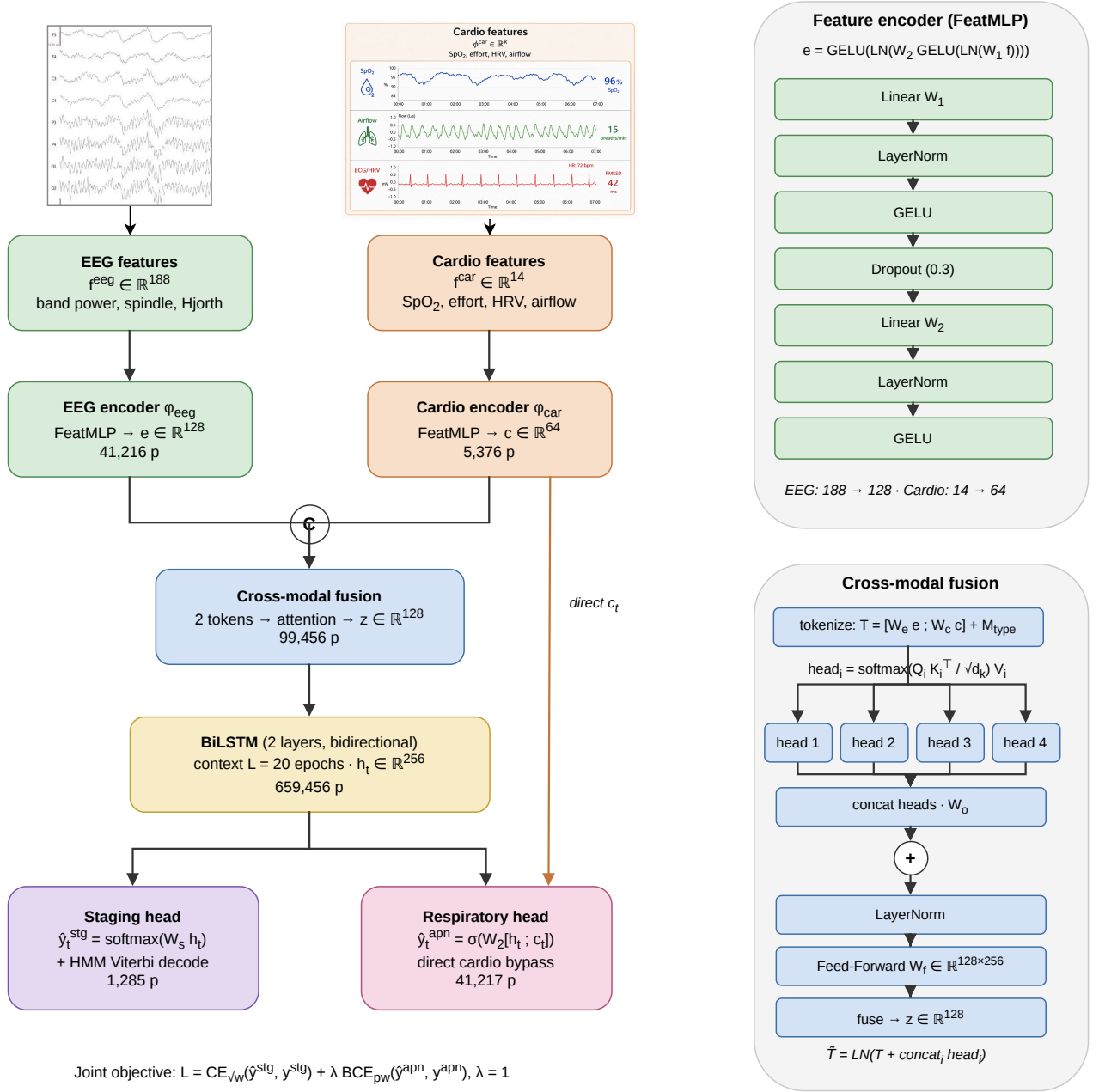


Fig. 2. The two-stream, multi-task model. (A) The neural features and the cardiorespiratory features are encoded in parallel, combined by a cross-modal fusion block, and read across the night by a bidirectional LSTM. A staging head reads the recurrent state; a respiratory head reads the recurrent state together with a direct copy of the cardiorespiratory embedding, so oxygen-desaturation and effort cues reach the respiratory decision without passing through the staging-oriented fusion. (B) Detail of the feature encoder and of the fusion block, which tokenizes the two modalities, runs four-head attention, and combines the result with a residual connection.

V. EXPERIMENTS

A. Protocol and metrics

All results use ten-fold, patient-independent cross-validation, so no patient appears in both training and test, and within each training split a disjoint set of patients is held out for early stopping. Staging is scored by accuracy, macro-F1 $\frac{1}{K} \sum_k F1_k$, and Cohen's κ . The respiratory task is scored by the area under the ROC curve (AUC) and average precision (AP), which suits the 16% event rate.

B. Compared Methods

We benchmark the model against a wide set of approaches on the same folds. The *deep* group holds convolutional and recurrent networks trained from scratch (DeepSleepNet, AttnSleep, a four-channel CNN with BiLSTM), a network transferred from Sleep-EDF, sequence CNN-BiLSTM models on four and seven channels, and a raw-signal convolutional network on all 14 channels. The *feature* group holds a single gradient-boosting classifier, a feature-sequence BiLSTM, a learnable refiner, a hemispheric graph-attention model with a state-space decoder, a stacking ensemble, and a four-member

TABLE II

SLEEP-STAGING BENCHMARK ON ISLEEPS, TEN-FOLD PATIENT-INDEPENDENT. ALL ROWS ARE THE FIVE-CLASS STAGING TASK AT THE HIDDEN-MARKOV OPERATING POINT WHERE AVAILABLE. CH. IS THE NUMBER OF INPUT CHANNELS; “FEAT” MARKS ENGINEERED FEATURES. BEST PER COLUMN IN **BOLD**. THE PROPOSED MODEL IS THE ONLY ONE THAT ALSO DETECTS RESPIRATORY EVENTS (TABLE IV).

Model	Group	Input	Params ↓	Acc ↑	Macro-F1 ↑	κ ↑
<i>Published single-channel baselines [13]</i>						
LSTM	recurrent	1 EEG	–	0.747	0.677	0.640
Transformer	attention	1 EEG	–	0.674	0.594	0.540
CNN-ResNet18	convolution	1 EEG	~11 M	0.617	0.544	0.480
<i>Deep models (this study)</i>						
DeepSleepNet [2]	conv+LSTM	1 EEG	~21 M	0.615	0.567	0.483
AttnSleep [5]	attention	1 EEG	~0.5 M	0.686	0.629	0.577
CNN+BiLSTM	conv+LSTM	4 EEG	~0.6 M	0.613	0.565	0.485
Sleep-EDF transfer	transfer	4 EEG	~0.6 M	0.623	0.582	0.488
Sequence CNN-BiLSTM	conv+LSTM	4 EEG	~0.8 M	0.645	0.609	0.526
Sequence CNN-BiLSTM	conv+LSTM	7 ch	~0.8 M	0.654	0.610	0.544
Raw multimodal CNN	conv+LSTM	14 ch	0.95 M	0.655	0.612	0.531
<i>Feature and ensemble models (this study)</i>						
Gradient boosting (single)	feature	7 ch feat	–	0.655	0.499	0.485
Feature-sequence BiLSTM	feature	7 ch feat	1.2 M	0.676	0.639	0.566
Learnable refiner	feature	7 ch feat	~0.5 M	0.704	0.661	0.600
Graph attention + SSM	feature	7 ch feat	0.49 M	0.730	0.703	0.633
Stacking ensemble	feature	7 ch feat	–	0.735	0.692	0.639
Gradient-boosting ensemble	feature	7 ch feat	–	0.746	0.675	0.641
<i>Proposed multimodal multi-task model</i>						
MM-Net (neural only)	feature	7 ch feat	0.77 M	0.721	0.635	0.603
MM-Net (neural + cardio)	feature	14 ch feat	0.86 M	0.721	0.645	0.606

gradient-boosting ensemble. The published single-channel baselines for the corpus are included for reference [13]. All numbers are ten-fold patient-independent, at the hidden-Markov operating point where a model provides one.

C. Staging Benchmark and Comparison

Table II shows a clear pattern. Every raw-signal deep network, whether trained from scratch or transferred, stays near 0.62 to 0.66 accuracy, and the strong feature and ensemble models cluster between 0.70 and 0.75. Staging on this stroke cohort saturates in that band regardless of architecture, which matches the concurrent finding that depth alone does not help here [16]. The proposed model stages at 0.721 accuracy, inside the top group and above every from-scratch deep network, while using under a million parameters. What sets it apart is not a staging record but a second output: it is the only model in the table that also detects respiratory events.

D. Per-class Staging Performance

Table III breaks staging down by stage. The pattern is the one this cohort imposes on every method: N2, Wake, and REM are recovered well, N3 is moderate, and N1 is the hard stage for all models, because N1 is transient and rare. The raw convolutional network reaches the highest N1 F1, which fits its sensitivity to short events, while the gradient-boosting ensemble leads on the stable stages. The proposed model tracks the strong feature models across every stage. Per-class scores were not recorded for some baselines and are marked with a dash.

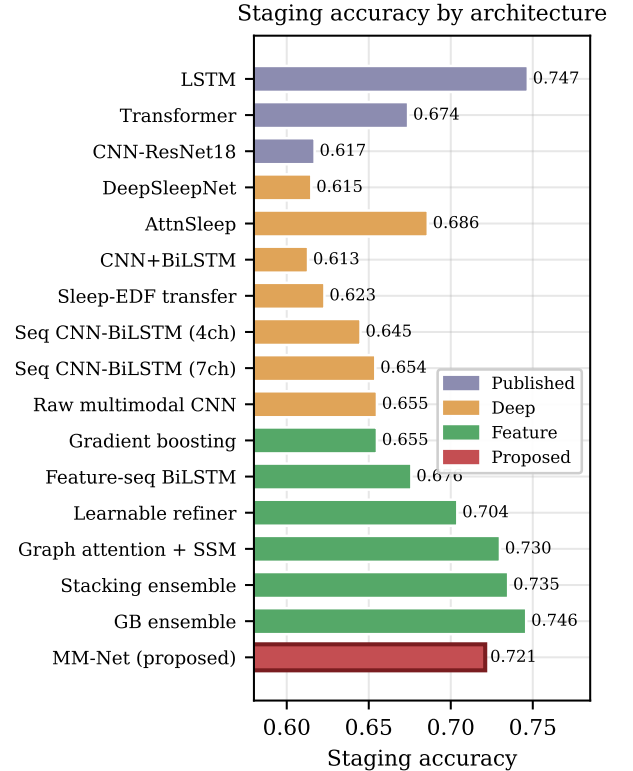


Fig. 3. Staging accuracy across the fifteen benchmarked architectures (ten-fold, patient-independent), grouped by family. The proposed model sits inside the top band of the feature and ensemble models while remaining compact.

E. Ablation Study: Effect of the Cardiorespiratory Stream

Table IV reports the proposed model in full and turns the cardiorespiratory stream on and off on the same folds. The

TABLE III

PER-CLASS STAGING F1 (WAKE, N1, N2, N3, REM) ACROSS THE COMPARED ARCHITECTURES, TEN-FOLD PATIENT-INDEPENDENT, AT THE HIDDEN-MARKOV OPERATING POINT. A DASH MARKS A MODEL FOR WHICH PER-CLASS SCORES WERE NOT RECORDED IN OUR RUNS. BEST PER COLUMN IN **BOLD**.

Model	Group	W ↑	N1 ↑	N2 ↑	N3 ↑	R ↑
<i>Published single-channel baselines</i> [13]						
LSTM	recurrent	—	—	—	—	—
Transformer	attention	—	—	—	—	—
CNN-ResNet18	convolution	—	—	—	—	—
<i>Deep models (this study)</i>						
DeepSleepNet [2]	conv+LSTM	—	—	—	—	—
AttnSleep [5]	attention	—	—	—	—	—
CNN + BiLSTM	conv+LSTM	—	—	—	—	—
Sleep-EDF transfer	transfer	—	—	—	—	—
Sequence CNN-BiLSTM (4 ch)	conv+LSTM	—	—	—	—	—
Sequence CNN-BiLSTM (7 ch)	conv+LSTM	—	—	—	—	—
Raw multimodal CNN	conv+LSTM	0.74	0.37	0.73	0.56	0.67
<i>Feature and ensemble models (this study)</i>						
Gradient boosting (single)	feature	0.74	0.02	0.72	0.56	0.46
Feature-sequence BiLSTM	feature	—	—	—	—	—
Learnable refiner	feature	—	—	—	—	—
Graph attention + SSM	feature	—	—	—	—	—
Stacking ensemble	feature	—	—	—	—	—
Gradient-boosting ensemble	feature	0.80	0.32	0.80	0.70	0.76
<i>Proposed multimodal multi-task model</i>						
MM-Net (neural only)	feature	0.77	0.24	0.78	0.68	0.70
MM-Net (neural + cardio)	feature	0.77	0.27	0.78	0.68	0.72

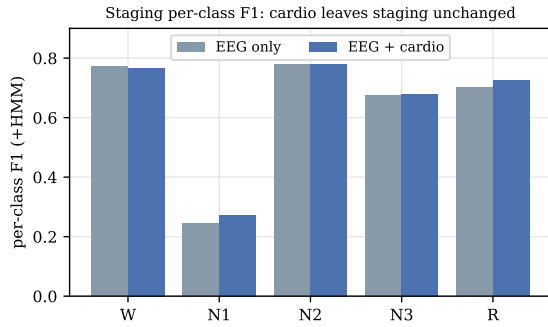


Fig. 4. Per-class staging F1 for the neural-only and the neural-plus-cardiorespiratory model. Adding the cardiorespiratory channels leaves every stage at its full level, including Wake and REM.

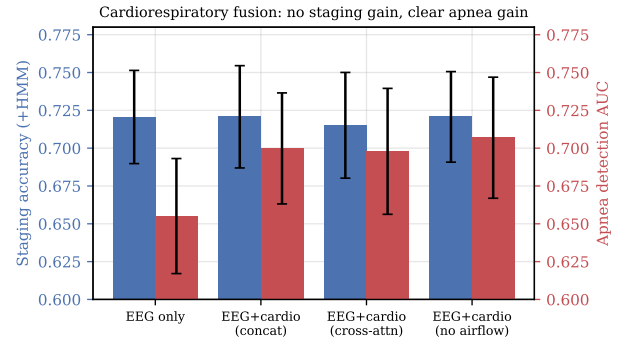


Fig. 5. Effect of the cardiorespiratory stream. Staging accuracy (blue) stays at its full level while respiratory-event AUC (red) rises sharply once the cardiorespiratory channels are read. Error bars are fold standard deviations.

model stages sleep at 0.721 accuracy and detects respiratory events at 0.700 AUC with 0.333 average precision. Removing the cardiorespiratory stream drops respiratory detection to 0.655 AUC, the most a model can do from cortical arousals alone, while staging is unchanged, because the neural channels carry staging and the breathing channels carry breathing. Attention and plain concatenation give the same numbers, so the gain is the modality. Per-class F1 confirms the division of labor: the cardiorespiratory stream leaves every stage at its full level, including Wake and REM, the stages where the multimodal literature predicts a change.

F. Physiological Validity of the Respiratory Read-out

Because the events were scored from airflow, a model that keyed on the airflow channel would be right for a trivial reason. We remove the two airflow features and test again. Detection

is not merely preserved, it is marginally higher, at 0.707 AUC (Table IV, last row). The model therefore does not lean on the labeling signal. It reads the fall in blood oxygen, the straining of the chest and abdomen, the beat-to-beat change in the heart, and the cortical arousal that ends the event. This is the useful regime, because it detects the bodily consequences of disordered breathing rather than repeating a measurement a technician has already made.

G. Model Complexity and Capability

Table V sets the proposed model beside representative alternatives on size and capability. A single compact network, under a million parameters and an order of magnitude smaller than the raw-signal transformers, delivers both a competitive stage and a respiratory read-out from one pass over the

TABLE IV

THE PROPOSED MULTIMODAL MULTI-TASK MODEL IN FULL, TEN-FOLD PATIENT-INDEPENDENT, AND THE EFFECT OF THE CARDIORESPIRATORY STREAM. STAGING IS AT THE HIDDEN-MARKOV OPERATING POINT; PER-CLASS VALUES ARE F1. THE RESPIRATORY TASK IS SCORED BY AUC AND AVERAGE PRECISION (AP). FOLD STANDARD DEVIATIONS IN PARENTHESES FOR THE HEADLINE METRICS.

Variant	Staging			Per-class F1 $\uparrow$					Respiratory	
	Acc $\uparrow$	mF1 $\uparrow$	$\kappa\uparrow$	W	N1	N2	N3	R	AUC $\uparrow$	AP $\uparrow$
Neural only	0.721 (.031)	0.635	0.603	0.77	0.24	0.78	0.68	0.70	0.655 (.038)	0.243
Neural + cardio (concat)	0.721 (.034)	0.645	0.606	0.77	0.27	0.78	0.68	0.72	0.700 (.037)	0.333
Neural + cardio (attention)	0.715 (.035)	0.636	0.600	0.76	0.25	0.78	0.66	0.72	0.698 (.042)	0.324
airflow removed	0.721 (.030)	0.649	0.609	0.77	0.29	0.78	0.67	0.74	0.707 (.040)	0.328

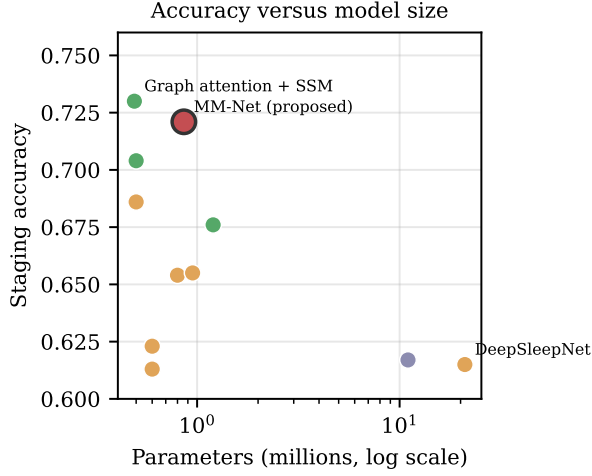


Fig. 6. Staging accuracy against model size. The proposed model reaches top-band accuracy with under a million parameters, an order of magnitude smaller than the raw-signal networks, and is the only point that also detects respiratory events.

TABLE V

MODEL SIZE AND CLINICAL CAPABILITY. THE PROPOSED MODEL IS COMPACT AND IS THE ONLY ONE THAT PRODUCES BOTH A STAGE AND A RESPIRATORY-EVENT LABEL.

Model	Input	Params $\downarrow$	Outputs
CNN-ResNet18 [13]	1 EEG	~ 11 M	stage
DeepSleepNet [2]	1 EEG	~ 21 M	stage
Feature-sequence BiLSTM	7 ch feat	1.2 M	stage
Graph attention + SSM	7 ch feat	0.49 M	stage
Raw multimodal CNN	14 ch	0.95 M	stage + apnea
MM-Net (proposed)	14 ch feat	0.86 M	stage + apnea

recording. None of the staging-only models, at any size, can supply the second output.

VI. DISCUSSION

The result is a single model that reads the whole sleeping body and speaks to two clinical questions at once, in the population where both matter. Across fifteen architectures, staging accuracy on this cohort saturates near 0.72 to 0.75, and the proposed model sits inside that top band while adding a respiratory capability the staging models lack. The respiratory read-out is not a by-product of the label. It survives deletion of the airflow channel and rests on oxygen desaturation, effort, and arousal, the same signs a physician reads at the bedside.

The study also draws a clean physiological line. The cortical channels carry sleep staging, and the cardiorespiratory channels carry breathing, and the model places each where it belongs rather than blurring them. That the neural and multimodal variants stage alike is not a shortfall; it is the correct outcome, and it tells us the second modality is adding a genuinely new capability rather than reshuffling the first. We view this separation as a strength for clinical use, because a reader can trust that the respiratory output reflects breathing and the staging output reflects the brain.

Two directions follow naturally. Event-level metrics, such as sensitivity per event and false alarms per hour, would sit beside the epoch-level AUC and speak more directly to a sleep technician, and finer modeling of the breathing waveforms is likely to raise the respiratory number further. Linking the multimodal representation to stroke severity and recovery is the larger opportunity, and one the cardiorespiratory signal is well placed to serve.

VII. CONCLUSION

We built a model that reads the full polysomnogram of stroke patients and, from one pass, stages their sleep and detects their respiratory events. Against a benchmark of fifteen architectures under patient-independent evaluation, it reaches 0.721 staging accuracy, in the top band, and 0.700 respiratory-event AUC, and the respiratory read-out holds when the airflow channel is removed, which shows it reads physiology rather than the label. Reading the whole recording, not the brain alone, gives the model a second clinical voice in the population that needs it most. Code and features are released.

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